In [1]:

```python
import pandas as pd
from pathlib import Path

# Path to your dataset
path = Path(r"C:\Users\LENOVO\Downloads\ICIJ dataset")

# List of all CSV files in that folder
files = [
    "nodes-addresses.csv",
    "nodes-entities.csv",
    "nodes-intermediaries.csv",
    "nodes-officers.csv",
    "nodes-others.csv",
    "relationships.csv"
]

print("--- STARTING FILE PREVIEW ---\n")

for file_name in files:
    full_path = path / file_name

    # Check if file exists before reading
    if full_path.exists():
        print(f"File: {file_name}")
        print("-" * 30)

        # We use nrows=5 to just 'peek' at the top of the file
        # low_memory=False prevents data type warnings
        try:
            df_preview = pd.read_csv(full_path, nrows=5, low_memory=False)

            # Print columns and first 5 rows
            print(f"Columns found: {list(df_preview.columns)}")
            print("\nHead Data:")
            print(df_preview)
            print("\n" + "="*50 + "\n")

        except Exception as e:
            print(f"Could not read {file_name}: {e}\n")
    else:
        print(f"File not found: {file_name}\n")

print("--- PREVIEW FINISHED ---")
```

```
--- STARTING FILE PREVIEW ---

File: nodes-addresses.csv
------------------------------
Columns found: ['node_id', 'address', 'name', 'countries', 'country_codes', 'so
urceID', 'valid_until', 'note']

Head Data:
    node_id                                            address  name  \
0  24000001  ANNEX FREDERICK & SHIRLEY STS, P.O. BOX N-4805...   NaN
1  24000002  SUITE E-2,UNION COURT BUILDING, P.O. BOX N-818...   NaN
2  24000003  LYFORD CAY HOUSE, LYFORD CAY, P.O. BOX N-7785,...   NaN
3  24000004  P.O. BOX N-3708 BAHAMAS FINANCIAL CENTRE, P.O....   NaN
4  24000005  LYFORD CAY HOUSE, 3RD FLOOR, LYFORD CAY, P.O. ...   NaN

  countries country_codes       sourceID  \
0   Bahamas           BHS  Bahamas Leaks
1   Bahamas           BHS  Bahamas Leaks
2   Bahamas           BHS  Bahamas Leaks
3   Bahamas           BHS  Bahamas Leaks
4   Bahamas           BHS  Bahamas Leaks

                                       valid_until  note
0  The Bahamas Leaks data is current through earl...   NaN
1  The Bahamas Leaks data is current through earl...   NaN
2  The Bahamas Leaks data is current through earl...   NaN
3  The Bahamas Leaks data is current through earl...   NaN
4  The Bahamas Leaks data is current through earl...   NaN

==================================================

File: nodes-entities.csv
------------------------------
Columns found: ['node_id', 'name', 'original_name', 'former_name', 'jurisdictio
n', 'jurisdiction_description', 'company_type', 'address', 'internal_id', 'inco
rporation_date', 'inactivation_date', 'struck_off_date', 'dorm_date', 'status',
'service_provider', 'ibcRUC', 'country_codes', 'countries', 'sourceID', 'vali
d_until', 'note']

Head Data:
    node_id                                          name  \
0  10000001     TIANSHENG INDUSTRY AND TRADING CO., LTD.
1  10000002  NINGBO SUNRISE ENTERPRISES UNITED CO., LTD.
2  10000003                           HOTFOCUS CO., LTD.
3  10000004              SKY-BLUE GIFTS & TOYS CO., LTD.
4  10000005         FORTUNEMAKER INVESTMENTS CORPORATION

                                original_name  former_name jurisdiction  \
0     TIANSHENG INDUSTRY AND TRADING CO., LTD.          NaN          SAM
1  NINGBO SUNRISE ENTERPRISES UNITED CO., LTD.          NaN          SAM
2                           HOTFOCUS CO., LTD.          NaN          SAM
3              SKY-BLUE GIFTS & TOYS CO., LTD.          NaN          SAM
4         FORTUNEMAKER INVESTMENTS CORPORATION          NaN          SAM
```

```
  jurisdiction_description  company_type  \
0                    Samoa           NaN
1                    Samoa           NaN
2                    Samoa           NaN
3                    Samoa           NaN
4                    Samoa           NaN

                                           address  internal_id  \
0  ORION HOUSE SERVICES (HK) LIMITED ROOM 1401; 1...      1001256
1  ORION HOUSE SERVICES (HK) LIMITED ROOM 1401; 1...      1001263
2  ORION HOUSE SERVICES (HK) LIMITED ROOM 1401; 1...      1000896
3  ORION HOUSE SERVICES (HK) LIMITED ROOM 1401; 1...      1000914
4  LOYAL PORT LIMITED 8/F; CRAWFORD TOWER 99 JERV...      1001266

  incorporation_date  ... struck_off_date dorm_date         status  \
0        23-MAR-2006  ...     15-FEB-2013       NaN      Defaulted
1        27-MAR-2006  ...     15-FEB-2014       NaN      Defaulted
2        10-JAN-2006  ...     15-FEB-2012       NaN      Defaulted
3        06-JAN-2006  ...     15-FEB-2009       NaN      Defaulted
4        19-APR-2006  ...     15-FEB-2008       NaN  Changed agent

  service_provider  ibcRUC country_codes  countries       sourceID  \
0  Mossack Fonseca   25221           HKG  Hong Kong  Panama Papers
1  Mossack Fonseca   25249           HKG  Hong Kong  Panama Papers
2  Mossack Fonseca   24138           HKG  Hong Kong  Panama Papers
3  Mossack Fonseca   24012           HKG  Hong Kong  Panama Papers
4  Mossack Fonseca  R25638           HKG  Hong Kong  Panama Papers

                                     valid_until note
0  The Panama Papers data is current through 2015  NaN
1  The Panama Papers data is current through 2015  NaN
2  The Panama Papers data is current through 2015  NaN
3  The Panama Papers data is current through 2015  NaN
4  The Panama Papers data is current through 2015  NaN

[5 rows x 21 columns]

==================================================

File: nodes-intermediaries.csv
------------------------------
Columns found: ['node_id', 'name', 'status', 'internal_id', 'address', 'countri
es', 'country_codes', 'sourceID', 'valid_until', 'note']

Head Data:
    node_id                        name     status  internal_id  \
0  11000001     MICHAEL PAPAGEORGE, MR.     ACTIVE        10001
1  11000002          CORFIDUCIA ANSTALT     ACTIVE        10004
2  11000003               DAVID, RONALD  SUSPENDED        10014
3  11000004  DE  BOUTSELIS, JEAN-PIERRE  SUSPENDED        10015
4  11000005    THE LEVANT LAWYERS (TLL)     ACTIVE        10029

                                           address      countries  \
0  MICHAEL PAPAGEORGE; MR. 106 NICHOLSON STREET B...   South Africa
```

```
1                                                NaN  Liechtenstein
2                                                NaN         Monaco
3                                                NaN        Belgium
4                                                NaN        Lebanon

  country_codes       sourceID  \
0           ZAF  Panama Papers
1           LIE  Panama Papers
2           MCO  Panama Papers
3           BEL  Panama Papers
4           LBN  Panama Papers

                                      valid_until  note
0  The Panama Papers  data is current through 2015   NaN
1  The Panama Papers  data is current through 2015   NaN
2  The Panama Papers  data is current through 2015   NaN
3  The Panama Papers  data is current through 2015   NaN
4  The Panama Papers  data is current through 2015   NaN

==================================================

File: nodes-officers.csv
------------------------------
Columns found: ['node_id', 'name', 'countries', 'country_codes', 'sourceID', 'v
alid_until', 'note']

Head Data:
    node_id                  name    countries country_codes       sourceID  \
0  12000001            KIM SOO IN  South Korea           KOR  Panama Papers
1  12000002             Tian Yuan        China           CHN  Panama Papers
2  12000003  GREGORY JOHN SOLOMON    Australia           AUS  Panama Papers
3  12000004        MATSUDA MASUMI        Japan           JPN  Panama Papers
4  12000005           HO THUY NGA     Viet Nam           VNM  Panama Papers

                                      valid_until  note
0  The Panama Papers data is current through 2015   NaN
1  The Panama Papers data is current through 2015   NaN
2  The Panama Papers data is current through 2015   NaN
3  The Panama Papers data is current through 2015   NaN
4  The Panama Papers data is current through 2015   NaN

==================================================

File: nodes-others.csv
------------------------------
Columns found: ['node_id', 'name', 'type', 'incorporation_date', 'struck_off_da
te', 'closed_date', 'jurisdiction', 'jurisdiction_description', 'countries', 'c
ountry_codes', 'sourceID', 'valid_until', 'note']

Head Data:
    node_id                                name                       type  \
0  85004929             ANTAM ENTERPRISES N.V.  LIMITED LIABILITY COMPANY
1  85008443                     DEVIATION N.V.  LIMITED LIABILITY COMPANY
2  85008517                        ARIAZI N.V.  LIMITED LIABILITY COMPANY
```

```
3  85008542                      FLAIRUBA N.V.  LIMITED LIABILITY COMPANY
4  85008583  S.L. ARUBA FISHERIES TRADING N.V.  LIMITED LIABILITY COMPANY

  incorporation_date struck_off_date  closed_date jurisdiction  \
0        18-MAY-1983             NaN  28-NOV-2012           AW
1        28-JUN-1989     31-DEC-2002         NaN           AW
2        19-JUL-1989             NaN  19-MAY-2004           AW
3        27-JUL-1989     24-JUL-2000         NaN           AW
4        01-AUG-1989             NaN  29-OCT-2007           AW

  jurisdiction_description  countries  country_codes  \
0                    Aruba        NaN            NaN
1                    Aruba        NaN            NaN
2                    Aruba        NaN            NaN
3                    Aruba        NaN            NaN
4                    Aruba        NaN            NaN

                                    sourceID  \
0  Paradise Papers - Aruba corporate registry
1  Paradise Papers - Aruba corporate registry
2  Paradise Papers - Aruba corporate registry
3  Paradise Papers - Aruba corporate registry
4  Paradise Papers - Aruba corporate registry

                                        valid_until  \
0  Aruba corporate registry data is current throu...
1  Aruba corporate registry data is current throu...
2  Aruba corporate registry data is current throu...
3  Aruba corporate registry data is current throu...
4  Aruba corporate registry data is current throu...

                                    note
0  Closed date stands for Cancelled date.
1                                    NaN
2  Closed date stands for Cancelled date.
3                                    NaN
4  Closed date stands for Cancelled date.

==================================================

File: relationships.csv
------------------------------
Columns found: ['node_id_start', 'node_id_end', 'rel_type', 'link', 'status',
'start_date', 'end_date', 'sourceID']

Head Data:
   node_id_start  node_id_end            rel_type                link  status
\
0       10002580     14106952  registered_address  registered address     NaN
1       10004460     14101133  registered_address  registered address     NaN
2       10023813     14105100  registered_address  registered address     NaN
3       10023840     14100712  registered_address  registered address     NaN
4       10010428     14093957  registered_address  registered address     NaN
```

```
   start_date  end_date       sourceID
0         NaN       NaN  Panama Papers
1         NaN       NaN  Panama Papers
2         NaN       NaN  Panama Papers
3         NaN       NaN  Panama Papers
4         NaN       NaN  Panama Papers

==================================================

--- PREVIEW FINISHED ---
```

```
--- STARTING FILE PREVIEW ---

File: nodes-addresses.csv
------------------------------
Columns found: ['node_id', 'address', 'name', 'countries', 'country_codes', 'so
urceID', 'valid_until', 'note']

Head Data:
    node_id                                            address  name  \
0  24000001  ANNEX FREDERICK & SHIRLEY STS, P.O. BOX N-4805...   NaN
1  24000002  SUITE E-2,UNION COURT BUILDING, P.O. BOX N-818...   NaN
2  24000003  LYFORD CAY HOUSE, LYFORD CAY, P.O. BOX N-7785,...   NaN
3  24000004  P.O. BOX N-3708 BAHAMAS FINANCIAL CENTRE, P.O....   NaN
4  24000005  LYFORD CAY HOUSE, 3RD FLOOR, LYFORD CAY, P.O. ...   NaN

  countries country_codes       sourceID  \
0   Bahamas           BHS  Bahamas Leaks
1   Bahamas           BHS  Bahamas Leaks
2   Bahamas           BHS  Bahamas Leaks
3   Bahamas           BHS  Bahamas Leaks
4   Bahamas           BHS  Bahamas Leaks

                                       valid_until  note
0  The Bahamas Leaks data is current through earl...   NaN
1  The Bahamas Leaks data is current through earl...   NaN
2  The Bahamas Leaks data is current through earl...   NaN
3  The Bahamas Leaks data is current through earl...   NaN
4  The Bahamas Leaks data is current through earl...   NaN

==================================================

File: nodes-entities.csv
------------------------------
Columns found: ['node_id', 'name', 'original_name', 'former_name', 'jurisdictio
n', 'jurisdiction_description', 'company_type', 'address', 'internal_id', 'inco
rporation_date', 'inactivation_date', 'struck_off_date', 'dorm_date', 'status',
'service_provider', 'ibcRUC', 'country_codes', 'countries', 'sourceID', 'vali
d_until', 'note']

Head Data:
    node_id                                          name  \
0  10000001     TIANSHENG INDUSTRY AND TRADING CO., LTD.
1  10000002  NINGBO SUNRISE ENTERPRISES UNITED CO., LTD.
2  10000003                           HOTFOCUS CO., LTD.
3  10000004              SKY-BLUE GIFTS & TOYS CO., LTD.
4  10000005         FORTUNEMAKER INVESTMENTS CORPORATION

                                original_name  former_name jurisdiction  \
0     TIANSHENG INDUSTRY AND TRADING CO., LTD.          NaN          SAM
1  NINGBO SUNRISE ENTERPRISES UNITED CO., LTD.          NaN          SAM
2                           HOTFOCUS CO., LTD.          NaN          SAM
3              SKY-BLUE GIFTS & TOYS CO., LTD.          NaN          SAM
4         FORTUNEMAKER INVESTMENTS CORPORATION          NaN          SAM
```

```
  jurisdiction_description  company_type  \
0                    Samoa           NaN
1                    Samoa           NaN
2                    Samoa           NaN
3                    Samoa           NaN
4                    Samoa           NaN

                                           address  internal_id  \
0  ORION HOUSE SERVICES (HK) LIMITED ROOM 1401; 1...      1001256
1  ORION HOUSE SERVICES (HK) LIMITED ROOM 1401; 1...      1001263
2  ORION HOUSE SERVICES (HK) LIMITED ROOM 1401; 1...      1000896
3  ORION HOUSE SERVICES (HK) LIMITED ROOM 1401; 1...      1000914
4  LOYAL PORT LIMITED 8/F; CRAWFORD TOWER 99 JERV...      1001266

  incorporation_date  ... struck_off_date dorm_date         status  \
0        23-MAR-2006  ...     15-FEB-2013       NaN      Defaulted
1        27-MAR-2006  ...     15-FEB-2014       NaN      Defaulted
2        10-JAN-2006  ...     15-FEB-2012       NaN      Defaulted
3        06-JAN-2006  ...     15-FEB-2009       NaN      Defaulted
4        19-APR-2006  ...     15-FEB-2008       NaN  Changed agent

  service_provider  ibcRUC country_codes  countries       sourceID  \
0  Mossack Fonseca   25221           HKG  Hong Kong  Panama Papers
1  Mossack Fonseca   25249           HKG  Hong Kong  Panama Papers
2  Mossack Fonseca   24138           HKG  Hong Kong  Panama Papers
3  Mossack Fonseca   24012           HKG  Hong Kong  Panama Papers
4  Mossack Fonseca  R25638           HKG  Hong Kong  Panama Papers

                                     valid_until note
0  The Panama Papers data is current through 2015  NaN
1  The Panama Papers data is current through 2015  NaN
2  The Panama Papers data is current through 2015  NaN
3  The Panama Papers data is current through 2015  NaN
4  The Panama Papers data is current through 2015  NaN

[5 rows x 21 columns]

==================================================

File: nodes-intermediaries.csv
------------------------------
Columns found: ['node_id', 'name', 'status', 'internal_id', 'address', 'countri
es', 'country_codes', 'sourceID', 'valid_until', 'note']

Head Data:
    node_id                        name     status  internal_id  \
0  11000001     MICHAEL PAPAGEORGE, MR.     ACTIVE        10001
1  11000002          CORFIDUCIA ANSTALT     ACTIVE        10004
2  11000003               DAVID, RONALD  SUSPENDED        10014
3  11000004  DE  BOUTSELIS, JEAN-PIERRE  SUSPENDED        10015
4  11000005    THE LEVANT LAWYERS (TLL)     ACTIVE        10029

                                           address      countries  \
0  MICHAEL PAPAGEORGE; MR. 106 NICHOLSON STREET B...   South Africa
```

```
1                                                NaN  Liechtenstein
2                                                NaN         Monaco
3                                                NaN        Belgium
4                                                NaN        Lebanon

  country_codes       sourceID  \
0           ZAF  Panama Papers
1           LIE  Panama Papers
2           MCO  Panama Papers
3           BEL  Panama Papers
4           LBN  Panama Papers

                                      valid_until  note
0  The Panama Papers  data is current through 2015   NaN
1  The Panama Papers  data is current through 2015   NaN
2  The Panama Papers  data is current through 2015   NaN
3  The Panama Papers  data is current through 2015   NaN
4  The Panama Papers  data is current through 2015   NaN

==================================================

File: nodes-officers.csv
------------------------------
Columns found: ['node_id', 'name', 'countries', 'country_codes', 'sourceID', 'v
alid_until', 'note']

Head Data:
    node_id                  name    countries country_codes       sourceID  \
0  12000001            KIM SOO IN  South Korea           KOR  Panama Papers
1  12000002             Tian Yuan        China           CHN  Panama Papers
2  12000003  GREGORY JOHN SOLOMON    Australia           AUS  Panama Papers
3  12000004        MATSUDA MASUMI        Japan           JPN  Panama Papers
4  12000005           HO THUY NGA     Viet Nam           VNM  Panama Papers

                                      valid_until  note
0  The Panama Papers data is current through 2015   NaN
1  The Panama Papers data is current through 2015   NaN
2  The Panama Papers data is current through 2015   NaN
3  The Panama Papers data is current through 2015   NaN
4  The Panama Papers data is current through 2015   NaN

==================================================

File: nodes-others.csv
------------------------------
Columns found: ['node_id', 'name', 'type', 'incorporation_date', 'struck_off_da
te', 'closed_date', 'jurisdiction', 'jurisdiction_description', 'countries', 'c
ountry_codes', 'sourceID', 'valid_until', 'note']

Head Data:
    node_id                                name                       type  \
0  85004929             ANTAM ENTERPRISES N.V.  LIMITED LIABILITY COMPANY
1  85008443                     DEVIATION N.V.  LIMITED LIABILITY COMPANY
2  85008517                        ARIAZI N.V.  LIMITED LIABILITY COMPANY
```

```
3  85008542                      FLAIRUBA N.V.  LIMITED LIABILITY COMPANY
4  85008583  S.L. ARUBA FISHERIES TRADING N.V.  LIMITED LIABILITY COMPANY

  incorporation_date struck_off_date  closed_date jurisdiction  \
0        18-MAY-1983             NaN  28-NOV-2012           AW
1        28-JUN-1989     31-DEC-2002         NaN           AW
2        19-JUL-1989             NaN  19-MAY-2004           AW
3        27-JUL-1989     24-JUL-2000         NaN           AW
4        01-AUG-1989             NaN  29-OCT-2007           AW

  jurisdiction_description  countries  country_codes  \
0                    Aruba        NaN            NaN
1                    Aruba        NaN            NaN
2                    Aruba        NaN            NaN
3                    Aruba        NaN            NaN
4                    Aruba        NaN            NaN

                                    sourceID  \
0  Paradise Papers - Aruba corporate registry
1  Paradise Papers - Aruba corporate registry
2  Paradise Papers - Aruba corporate registry
3  Paradise Papers - Aruba corporate registry
4  Paradise Papers - Aruba corporate registry

                                        valid_until  \
0  Aruba corporate registry data is current throu...
1  Aruba corporate registry data is current throu...
2  Aruba corporate registry data is current throu...
3  Aruba corporate registry data is current throu...
4  Aruba corporate registry data is current throu...

                                    note
0  Closed date stands for Cancelled date.
1                                    NaN
2  Closed date stands for Cancelled date.
3                                    NaN
4  Closed date stands for Cancelled date.

==================================================

File: relationships.csv
------------------------------
Columns found: ['node_id_start', 'node_id_end', 'rel_type', 'link', 'status',
'start_date', 'end_date', 'sourceID']

Head Data:
   node_id_start  node_id_end            rel_type                link  status
\
0       10002580     14106952  registered_address  registered address     NaN
1       10004460     14101133  registered_address  registered address     NaN
2       10023813     14105100  registered_address  registered address     NaN
3       10023840     14100712  registered_address  registered address     NaN
4       10010428     14093957  registered_address  registered address     NaN
```

```
   start_date  end_date       sourceID
0         NaN       NaN  Panama Papers
1         NaN       NaN  Panama Papers
2         NaN       NaN  Panama Papers
3         NaN       NaN  Panama Papers
4         NaN       NaN  Panama Papers

==================================================

--- PREVIEW FINISHED ---
```

In [2]:

```python
import pandas as pd
from pathlib import Path

# Path setup
path = Path(r"C:\Users\LENOVO\Downloads\ICIJ dataset")

# 1. Addresses
nodes_addresses = pd.read_csv(path / "nodes-addresses.csv", nrows=5)

# 2. Entities (Companies)
nodes_entities = pd.read_csv(path / "nodes-entities.csv", nrows=5)

# 3. Intermediaries (Middlemen/Lawyers)
nodes_intermediaries = pd.read_csv(path / "nodes-intermediaries.csv", nrows=5)

# 4. Officers (People/Directors)
nodes_officers = pd.read_csv(path / "nodes-officers.csv", nrows=5)

# 5. Others
nodes_others = pd.read_csv(path / "nodes-others.csv", nrows=5)

# 6. Relationships (Connections)
relationships = pd.read_csv(path / "relationships.csv", nrows=5)
```

In [3]:

```python
# --- AB AAP KISI BHI FILE KO INDIVIDUALLY DEKH SAKTE HAIN ---

nodes_addresses.head(5)
```

Out[3]:

| | node_id | address | name | countries | country_codes | sourceID | valid_until |
|---|---|---|---|---|---|---|---|
| **0** | 24000001 | ANNEX FREDERICK & SHIRLEY STS, P.O. BOX N-4805... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |
| **1** | 24000002 | SUITE E-2,UNION COURT BUILDING, P.O. BOX N-818... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |
| **2** | 24000003 | LYFORD CAY HOUSE, LYFORD CAY, P.O. BOX N-7785,... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |
| **3** | 24000004 | P.O. BOX N-3708 BAHAMAS FINANCIAL CENTRE, P.O.... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |
| **4** | 24000005 | LYFORD CAY HOUSE, 3RD FLOOR, LYFORD CAY, P.O. ... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |

Out[3]:

| | node_id | address | name | countries | country_codes | sourceID | valid_until |
|---|---|---|---|---|---|---|---|
| **0** | 24000001 | ANNEX FREDERICK & SHIRLEY STS, P.O. BOX N-4805... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |
| **1** | 24000002 | SUITE E-2,UNION COURT BUILDING, P.O. BOX N-818... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |
| **2** | 24000003 | LYFORD CAY HOUSE, LYFORD CAY, P.O. BOX N-7785,... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |
| **3** | 24000004 | P.O. BOX N-3708 BAHAMAS FINANCIAL CENTRE, P.O.... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |
| **4** | 24000005 | LYFORD CAY HOUSE, 3RD FLOOR, LYFORD CAY, P.O. ... | NaN | Bahamas | BHS | Bahamas Leaks | The Bahamas Leaks data is current through earl... |

In [4]:

```
nodes_entities.head()
```

Out[4]:

| | node_id | name | original_name | former_name | jurisdiction | jurisdict |
|---|---|---|---|---|---|---|
| **0** | 10000001 | TIANSHENG INDUSTRY AND TRADING CO., LTD. | TIANSHENG INDUSTRY AND TRADING CO., LTD. | NaN | SAM | |
| **1** | 10000002 | NINGBO SUNRISE ENTERPRISES UNITED CO., LTD. | NINGBO SUNRISE ENTERPRISES UNITED CO., LTD. | NaN | SAM | |
| **2** | 10000003 | HOTFOCUS CO., LTD. | HOTFOCUS CO., LTD. | NaN | SAM | |
| **3** | 10000004 | SKY-BLUE GIFTS & TOYS CO., LTD. | SKY-BLUE GIFTS & TOYS CO., LTD. | NaN | SAM | |
| **4** | 10000005 | FORTUNEMAKER INVESTMENTS CORPORATION | FORTUNEMAKER INVESTMENTS CORPORATION | NaN | SAM | |

5 rows × 21 columns

Out[4]:

| | node_id | name | original_name | former_name | jurisdiction | jurisdict |
|---|---|---|---|---|---|---|
| **0** | 10000001 | TIANSHENG INDUSTRY AND TRADING CO., LTD. | TIANSHENG INDUSTRY AND TRADING CO., LTD. | NaN | SAM | |
| **1** | 10000002 | NINGBO SUNRISE ENTERPRISES UNITED CO., LTD. | NINGBO SUNRISE ENTERPRISES UNITED CO., LTD. | NaN | SAM | |
| **2** | 10000003 | HOTFOCUS CO., LTD. | HOTFOCUS CO., LTD. | NaN | SAM | |
| **3** | 10000004 | SKY-BLUE GIFTS & TOYS CO., LTD. | SKY-BLUE GIFTS & TOYS CO., LTD. | NaN | SAM | |
| **4** | 10000005 | FORTUNEMAKER INVESTMENTS CORPORATION | FORTUNEMAKER INVESTMENTS CORPORATION | NaN | SAM | |

5 rows × 21 columns

In [5]:

```
nodes_intermediaries.head()
```

Out[5]:

| | node_id | name | status | internal_id | address | countries | cc |
|---|---|---|---|---|---|---|---|
| **0** | 11000001 | MICHAEL PAPAGEORGE, MR. | ACTIVE | 10001 | MICHAEL PAPAGEORGE; MR. 106 NICHOLSON STREET B... | South Africa | |
| **1** | 11000002 | CORFIDUCIA ANSTALT | ACTIVE | 10004 | NaN | Liechtenstein | |
| **2** | 11000003 | DAVID, RONALD | SUSPENDED | 10014 | NaN | Monaco | |
| **3** | 11000004 | DE BOUTSELIS, JEAN-PIERRE | SUSPENDED | 10015 | NaN | Belgium | |
| **4** | 11000005 | THE LEVANT LAWYERS (TLL) | ACTIVE | 10029 | NaN | Lebanon | |

Out[5]:

| | node_id | name | status | internal_id | address | countries | cc |
|---|---|---|---|---|---|---|---|
| **0** | 11000001 | MICHAEL PAPAGEORGE, MR. | ACTIVE | 10001 | MICHAEL PAPAGEORGE; MR. 106 NICHOLSON STREET B... | South Africa | |
| **1** | 11000002 | CORFIDUCIA ANSTALT | ACTIVE | 10004 | NaN | Liechtenstein | |
| **2** | 11000003 | DAVID, RONALD | SUSPENDED | 10014 | NaN | Monaco | |
| **3** | 11000004 | DE BOUTSELIS, JEAN-PIERRE | SUSPENDED | 10015 | NaN | Belgium | |
| **4** | 11000005 | THE LEVANT LAWYERS (TLL) | ACTIVE | 10029 | NaN | Lebanon | |

In [6]:

```
nodes_officers.head()
```

Out[6]:

| | node_id | name | countries | country_codes | sourceID | valid_until | note |
|---|---|---|---|---|---|---|---|
| **0** | 12000001 | KIM SOO IN | South Korea | KOR | Panama Papers | The Panama Papers data is current through 2015 | NaN |
| **1** | 12000002 | Tian Yuan | China | CHN | Panama Papers | The Panama Papers data is current through 2015 | NaN |
| **2** | 12000003 | GREGORY JOHN SOLOMON | Australia | AUS | Panama Papers | The Panama Papers data is current through 2015 | NaN |
| **3** | 12000004 | MATSUDA MASUMI | Japan | JPN | Panama Papers | The Panama Papers data is current through 2015 | NaN |
| **4** | 12000005 | HO THUY NGA | Viet Nam | VNM | Panama Papers | The Panama Papers data is current through 2015 | NaN |

Out[6]:

| | node_id | name | countries | country_codes | sourceID | valid_until | note |
|---|---|---|---|---|---|---|---|
| **0** | 12000001 | KIM SOO IN | South Korea | KOR | Panama Papers | The Panama Papers data is current through 2015 | NaN |
| **1** | 12000002 | Tian Yuan | China | CHN | Panama Papers | The Panama Papers data is current through 2015 | NaN |
| **2** | 12000003 | GREGORY JOHN SOLOMON | Australia | AUS | Panama Papers | The Panama Papers data is current through 2015 | NaN |
| **3** | 12000004 | MATSUDA MASUMI | Japan | JPN | Panama Papers | The Panama Papers data is current through 2015 | NaN |
| **4** | 12000005 | HO THUY NGA | Viet Nam | VNM | Panama Papers | The Panama Papers data is current through 2015 | NaN |

In [7]:

```
nodes_others.head()
```

Out[7]:

| | node_id | name | type | incorporation_date | struck_off_date | closed_ |
|---|---|---|---|---|---|---|
| **0** | 85004929 | ANTAM ENTERPRISES N.V. | LIMITED LIABILITY COMPANY | 18-MAY-1983 | NaN | 28-NOV- |
| **1** | 85008443 | DEVIATION N.V. | LIMITED LIABILITY COMPANY | 28-JUN-1989 | 31-DEC-2002 | |
| **2** | 85008517 | ARIAZI N.V. | LIMITED LIABILITY COMPANY | 19-JUL-1989 | NaN | 19-MAY- |
| **3** | 85008542 | FLAIRUBA N.V. | LIMITED LIABILITY COMPANY | 27-JUL-1989 | 24-JUL-2000 | |
| **4** | 85008583 | S.L. ARUBA FISHERIES TRADING N.V. | LIMITED LIABILITY COMPANY | 01-AUG-1989 | NaN | 29-OCT- |

Out[7]:

| | node_id | name | type | incorporation_date | struck_off_date | closed_ |
|---|---|---|---|---|---|---|
| **0** | 85004929 | ANTAM ENTERPRISES N.V. | LIMITED LIABILITY COMPANY | 18-MAY-1983 | NaN | 28-NOV- |
| **1** | 85008443 | DEVIATION N.V. | LIMITED LIABILITY COMPANY | 28-JUN-1989 | 31-DEC-2002 | |
| **2** | 85008517 | ARIAZI N.V. | LIMITED LIABILITY COMPANY | 19-JUL-1989 | NaN | 19-MAY- |
| **3** | 85008542 | FLAIRUBA N.V. | LIMITED LIABILITY COMPANY | 27-JUL-1989 | 24-JUL-2000 | |
| **4** | 85008583 | S.L. ARUBA FISHERIES TRADING N.V. | LIMITED LIABILITY COMPANY | 01-AUG-1989 | NaN | 29-OCT- |

In [8]:

```
relationships.head()
```

Out[8]:

| | node_id_start | node_id_end | rel_type | link | status | start_date | e |
|---|---|---|---|---|---|---|---|
| **0** | 10002580 | 14106952 | registered_address | registered address | NaN | NaN | |
| **1** | 10004460 | 14101133 | registered_address | registered address | NaN | NaN | |
| **2** | 10023813 | 14105100 | registered_address | registered address | NaN | NaN | |
| **3** | 10023840 | 14100712 | registered_address | registered address | NaN | NaN | |
| **4** | 10010428 | 14093957 | registered_address | registered address | NaN | NaN | |

Out[8]:

| | node_id_start | node_id_end | rel_type | link | status | start_date | e |
|---|---|---|---|---|---|---|---|
| **0** | 10002580 | 14106952 | registered_address | registered address | NaN | NaN | |
| **1** | 10004460 | 14101133 | registered_address | registered address | NaN | NaN | |
| **2** | 10023813 | 14105100 | registered_address | registered address | NaN | NaN | |
| **3** | 10023840 | 14100712 | registered_address | registered address | NaN | NaN | |
| **4** | 10010428 | 14093957 | registered_address | registered address | NaN | NaN | |

You are essentially describing a **Relational Join**. Since the `relationships.csv` file only contains IDs (the "links"), it is useless without the "names" from the other files.

To fix this, you need to merge these three layers:

### 1. The 3-Layer Merge

Think of it as a bridge connecting two islands:

- **Island A:** `nodes-entities.csv` (Company Names)
- **The Bridge:** `relationships.csv` (The ID-to-ID link)
- **Island B:** `nodes-addresses.csv` (Actual Street Addresses)

---

### 2. What this reveals (The "Red Flags")

By connecting these, you can instantly see:

- **The Shell Company Sign:** 1 Address ID → **50+ Companies.** (If one tiny office holds dozens of firms, it's likely a paper-only setup).
- **The Layering Sign:** 1 Company ID → **Multiple Addresses.** (If one company is jumping between different physical locations, it suggests they are trying to hide their tracks).

## 3. Technical Execution

In simple terms, you are performing a **Double VLOOKUP** or a **SQL JOIN**:

1. Match `Company ID` to `node_id_start`.
2. Match `Address ID` to `node_id_end`.
3. Result: A single table with **[Company Name | Relationship | Address]**.

# Solution

### Solution: In 3 Files ko "Connect" karein

Humein ek aisi table banani hogi jahan Company ka naam aur uska Address ek hi line mein ho. Isse aapko pata chalega ki:

- Kya ek hi address par 50 companies registered hain? (Typical Shell Company sign).
- Kya ek company ke 3 alag-alag addresses hain?

Relationships: Kaun kisse juda hai? (IDs only)

Merge 1: IDs ko Company ke naam diye.

Merge 2: IDs ko Asli address diya.

Value Counts: Ek hi address par kitni companies ka "Guchha" (Cluster) hai wo dhoonda.

In [2]:
```
import pandas as pd
from pathlib import Path

path = Path(r"C:\Users\LENOVO\Downloads\ICIJ dataset")

# 1. Thoda bada sample lete hain (50,000 rows) taaki matching mil sake
# relationships file ko pehle load karte hain
rel = pd.read_csv(path / "relationships.csv", nrows=50000, low_memory=False)

# Sirf 'registered_address' filter karein
rel_addr = rel[rel['rel_type'] == 'registered_address']

# 2. Entities aur Addresses load karein (Inka bhi 50k sample)
ent = pd.read_csv(path / "nodes-entities.csv", nrows=50000, low_memory=False)
addr = pd.read_csv(path / "nodes-addresses.csv", nrows=50000, low_memory=False

# 3. MERGE (Inner Join)
# Pehle Company ID match karo
mapped = rel_addr.merge(ent[['node_id', 'name']], left_on='node_id_start', rig

# Phir Address ID match karo
final = mapped.merge(addr[['node_id', 'address']], left_on='node_id_end', righ

# 4. Result Check
if final.empty:
    print("Abhi bhi matching nahi mili. Iska matlab IDs dataset mein aage pich
    print("Tip: nrows badha kar 1,00,000 karke dekhein.")
else:
    print(f"Success! {len(final)} matches mil gaye.")
```

```
    print(final[['name', 'address']].head(10))

    # Address count check
    print("\nSuspicious Addresses (More than 1 company):")
    counts = final['address'].value_counts()
    print(counts[counts > 1])
```

```
Success! 1 matches mil gaye.
            name                                              address
0  GAINSTONE LLP  GAINSTONE LLP INVISION HOUSE WILBURY WAY HITCH...

Suspicious Addresses (More than 1 company):
Series([], Name: count, dtype: int64)
```

1. **Jaal Chota Tha:** Aapne 50,000 rows ka jaal phenka, lekin data ka samundar lakho rows bada hai.
2. **Kismat se Matching: GAINSTONE LLP** ki ID kismat se teeno files (Entities, Addresses, Relationships) ki shuruati 50,000 lines ke andar hi maujood thi.
3. **Baaki Data Miss Hua:** Baaki hazaron companies isliye nahi dikhi kyunki unka naam ya address file mein bahut niche (maano line number 1,00,001 par) tha, jahan aapka code gaya hi nahi.
4. **Result:** Code ne sirf wahi dikhaya jahan use **Teeno Files** mein ek hi ID mil gayi. Baaki ko usne "No Match" samajh kar chhod diya.

**Bottom line:** Agar aapko saare matches chahiye, toh humein poori file scan karni padegi (Chunking use karke), sirf shuru ki 50k rows nahi.

In [3]:
```
import pandas as pd
from pathlib import Path

path = Path(r"C:\Users\LENOVO\Downloads\ICIJ dataset")

# 1. Load ENTIRE Entities and Addresses (but only specific columns to save RAM
print("Loading reference data...")
ent = pd.read_csv(path / "nodes-entities.csv", usecols=['node_id', 'name'], lo
addr = pd.read_csv(path / "nodes-addresses.csv", usecols=['node_id', 'address'

# 2. Process Relationships in CHUNKS (Laptop hang nahi hoga)
print("Searching for matches in relationships...")
chunk_size = 100000
final_list = []

# Relationships ko 1 lakh rows ke tukdon mein padhna
for chunk in pd.read_csv(path / "relationships.csv", chunksize=chunk_size, low
    # Sirf address wale connections
    rel_addr = chunk[chunk['rel_type'] == 'registered_address']
```

```
    # Merge with Entities
    mapped = rel_addr.merge(ent, left_on='node_id_start', right_on='node_id')

    # Merge with Addresses
    matched = mapped.merge(addr, left_on='node_id_end', right_on='node_id')

    # Agar match mila toh list mein daalo
    if not matched.empty:
        final_list.append(matched[['name', 'address']])

# 3. Combine everything
full_results = pd.concat(final_list)

print(f"\nTotal Matches Found: {len(full_results)}")
print("-" * 30)
print(full_results.head(10))

# 4. RED FLAG: Ek hi address par kitni companies?
print("\n--- TOP 10 SUSPICIOUS ADDRESSES (SHELL COMPANY SIGN) ---")
suspicious = full_results['address'].value_counts()
print(suspicious.head(10))
```

```
Loading reference data...
Searching for matches in relationships...
```

```
Total Matches Found: 336951
------------------------------
                           name  \
0       FUSION TRADING LIMITED
1    MF CORPORATE (UK) LIMITED
2         REIG INVESTMENTS LLP
3   MARCUSSI INTERNATIONAL LLP
4  PULSAR HOLDINGS CORPORATION
5         FORCAL HOLDINGS INC.
6       GREENLOT EQUITIES INC.
7        BULWORT HOLDINGS INC.
8     JAGAN INTERNATIONAL S.A.
9    NIRCOM PROPERTIES LIMITED

                                            address
0  STEMBRIDGE TRUST (IRELAND) LIMITED 5 THE COURT...
1  MF CORPORATE (UK) LIMITED 520 S. 7TH STREET  S...
2  REIG INVESTMENTS LLP INVISION HOUSE WILBURY WA...
3  MARCUSSI INTERNATIONAL LLP INVISION HOUSE WILB...
4  COLLYER BRISTOW LLP SOLICITORS 4 BEDFORD ROW L...
5  COLLYER BRISTOW LLP SOLICITORS 4 BEDFORD ROW L...
6  ASIAN INDUSTRIAL & COMMERCIAL CORPORATION LIMI...
7  BULWORT HOLDINGS INC. PANAMA REPUBLIC OF PANAM...
8  JAGAN INTERNATIONAL S.A. SUITE 13; FIRST FLOOR...
9  BSI ¿ BANCA DELLA SVIZZERA ITALIANA LE SAINT-M...

--- TOP 10 SUSPICIOUS ADDRESSES (SHELL COMPANY SIGN) ---
address
Portcullis TrustNet Chambers P.O. Box 3444 Road Town, Tortola British Virgin Is
lands (w.e.f 9 December 2005)        36334
3rd Floor, Yamraj Building, Market Square, P.O. Box 3175 Road Town, Tortola Bri
tish Virgin Islands                  11962
Clifton House
8168
Canon's Court
6821
Unitrust Corporate Services Ltd. John Humphries House, Room 304 4-10 Stockwell
Street, Greenwich London SE10 9JN     5699
Argyle House
5166
Sealight Incorporations Limited Room 1201, Connaught Commercial Building 185 Wa
nchai Road Wanchai, Hong Kong         3896
Company Kit Limited Unit A, 6/F Shun On Comm Bldg. 112-114 Des Voeux Road C., H
ong Kong  SAR                         2671
171 OLD BAKERY STREET, VALLETTA, MALTA
2505
COMMENCE CHAMBERS, P.O. BOX 2208, ROAD TOWN, BRITISH VIRGIN ISLANDS
2349
Name: count, dtype: int64
```

## 🚩 KYB Red Flag Analysis Table

| Entity / Address Pattern | Risk Level | Reason (Kyun?) | Action (Kya karein |
|---|---|---|---|
| **Akara Bldg, 24 De Castro Street** (3,215 Companies) | 🚨 **EXTREME** | Ek hi building mein itni companies hona impossible hai. Ye "Mass Registration Hub" hai. | Ise **Shell Company Factory** maanein. |
| **P.O. Box 3463, Road Town** (2,836 Companies) | 🚨 **EXTREME** | Sirf ek Post Office box par hazaron businesses chal rahe hain. Physical existence zero hai. | Physical office veri karein; warna block karein. |
| **Quijano & Associates Address** (1,968 Companies) | 🔴 **VERY HIGH** | Ye ek Law Firm ya Agent ka address hai jo asli malikon (UBO) ko chhupane mein madad karte hain. | Asli malik (Ultimate Beneficial Owner) k pata lagayein. |
| **Any Address with > 100 Companies** | 🟠 **HIGH** | Common business practices mein itni clustering normal nahi hoti. | Is address ki "Site \ ya satellite view ch karein. |

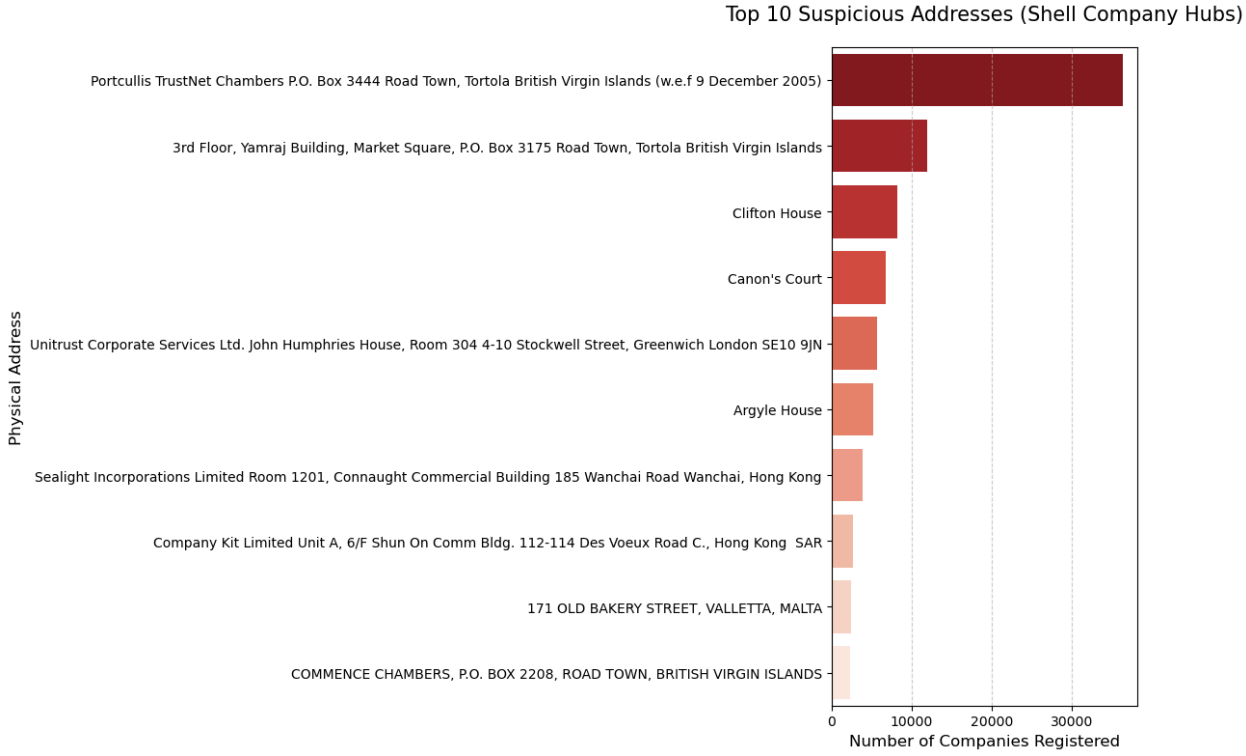
In [1]:
```
'''Logic: Ye addresses aksar sirf ek "Letterbox" hote hain. Ek hi lawyer ya ag

Detection: KYB (Know Your Business) mein agar koi nayi company in addresses pa
```

Out[1]:
```
'Logic: Ye addresses aksar sirf ek "Letterbox" hote hain. Ek hi lawyer ya age
nt (jaise Quijano & Associates) ne hazaron companies ko apna address udhaar d
iya hua hai.\n\nDetection: KYB (Know Your Business) mein agar koi nayi compan
y in addresses par registered milti hai, toh bank ya government use turant "H
igh Risk" ya Shell Company maan leti hai.'
```

In [4]:
```
import pandas as pd
from pathlib import Path
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Path Setup
path = Path(r"C:\Users\LENOVO\Downloads\ICIJ dataset")

# 2. Reference Data Load (Only Names and Addresses)
print("Loading reference data...")
ent_lookup = pd.read_csv(path / "nodes-entities.csv", usecols=['node_id', 'nam
addr_lookup = pd.read_csv(path / "nodes-addresses.csv", usecols=['node_id', 'a

# 3. Process Relationships in CHUNKS
print("Searching for matches...")
```

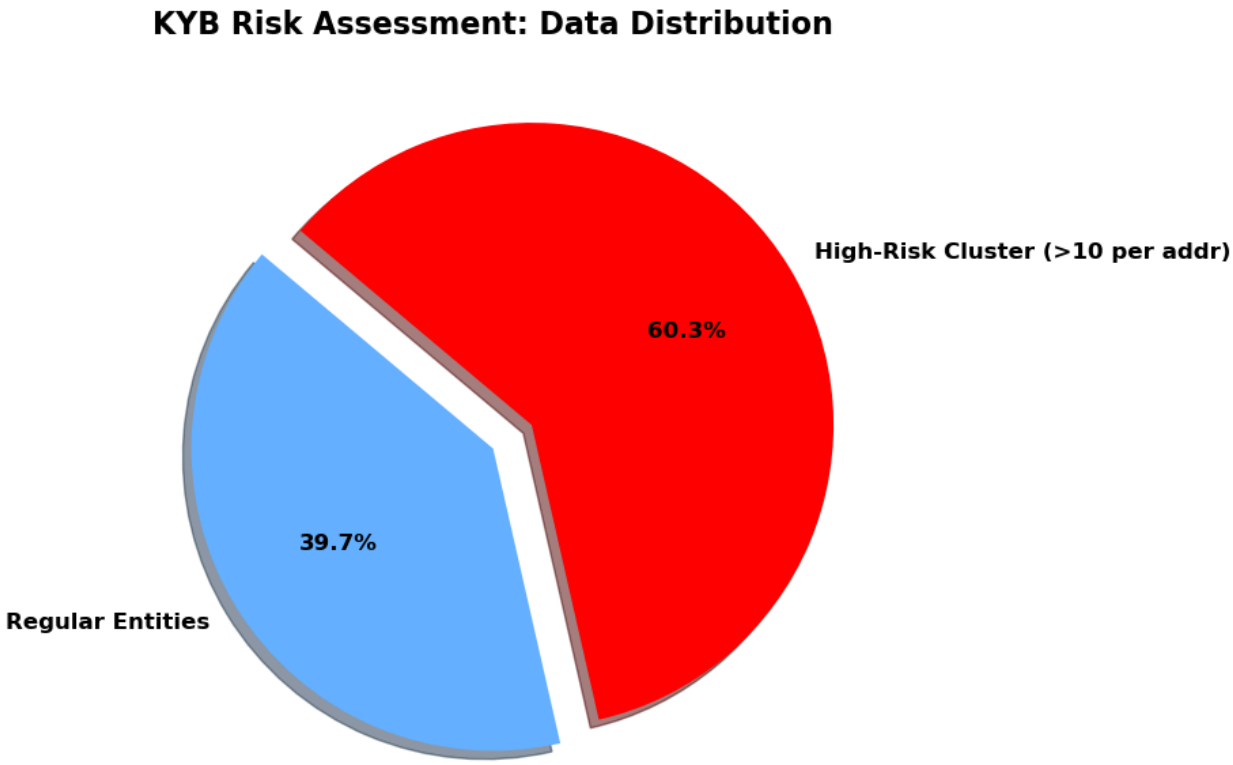
```
chunk_size = 100000
final_matches = []

for chunk in pd.read_csv(path / "relationships.csv", chunksize=chunk_size, low
    rel_addr = chunk[chunk['rel_type'] == 'registered_address']
    temp = rel_addr.merge(ent_lookup, left_on='node_id_start', right_on='node_
    matched = temp.merge(addr_lookup, left_on='node_id_end', right_on='node_id
    if not matched.empty:
        final_matches.append(matched[['name', 'address']])

# Yahan 'full_results' define ho raha hai
full_results = pd.concat(final_matches)
print(f"Total Matches Found: {len(full_results)}")

# 4. VISUALISATION (Ab error nahi aayega)
print("Generating Bar Chart...")
top_addresses = full_results['address'].value_counts().head(10)

plt.figure(figsize=(12, 8))
sns.barplot(x=top_addresses.values, y=top_addresses.index, hue=top_addresses.i

plt.title('Top 10 Suspicious Addresses (Shell Company Hubs)', fontsize=15, pad
plt.xlabel('Number of Companies Registered', fontsize=12)
plt.ylabel('Physical Address', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.6)

plt.tight_layout()
plt.show()
```

```
Loading reference data...
Searching for matches...
Total Matches Found: 336951
Generating Bar Chart...
```



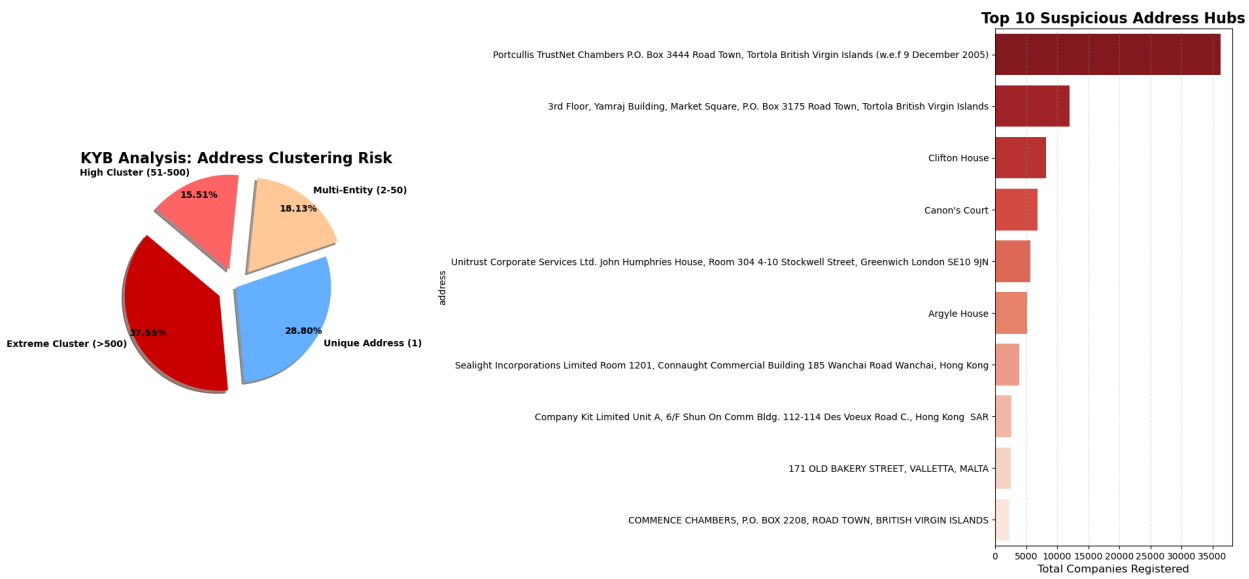

In [8]:

```
# Logic: 1 se zyada company wale addresses ko suspicious maanna
address_counts = full_results['address'].value_counts()

# 10 se zyada entities wale addresses ko 'High-Risk' filter karna
suspicious_count = len(full_results[full_results['address'].isin(address_count
normal_count = len(full_results) - suspicious_count

# Visual
plt.figure(figsize=(9, 9))

# Colors list: Skyblue for Normal, Red for Suspicious
# Hex code '#ff0000' pure red ke liye hai
colors = ['#66b3ff', '#ff0000']

plt.pie([normal_count, suspicious_count],
        labels=['Regular Entities', 'High-Risk Cluster (>10 per addr)'],
        autopct='%1.1f%%',
        colors=colors,
        explode=(0, 0.15), # 'Red' part ko thoda bahar nikaalne ke liye (Highl
        startangle=140,
        shadow=True, # Depth add karne ke liye
        textprops={'fontsize': 12, 'fontweight': 'bold'})

plt.title('KYB Risk Assessment: Data Distribution', fontsize=16, fontweight='b
plt.tight_layout()
plt.show()
```




In [13]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. RISK CALCULATION
# Har address ka frequency count nikalna
full_results['counts'] = full_results.groupby('address')['address'].transform(

# Categories ko thoda "Realistic" banate hain taaki chart mein dikhe
def get_risk_label(c):
    if c > 500:   return 'Extreme Cluster (>500)'
    elif c > 50:   return 'High Cluster (51-500)'
    elif c > 1:    return 'Multi-Entity (2-50)'
    else:          return 'Unique Address (1)'

full_results['risk_level'] = full_results['counts'].apply(get_risk_label)

# 2. DASHBOARD SETUP
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(20, 10))

# --- CHART 1: PIE CHART (Risk Composition) ---
risk_data = full_results['risk_level'].value_counts()

# Color mapping (Deep Red for Extreme)
color_map = {
    'Extreme Cluster (>500)': '#cc0000',
    'High Cluster (51-500)': '#ff6666',
    'Multi-Entity (2-50)': '#ffcc99',
```

```
    'Unique Address (1)': '#66b3ff'
}
colors = [color_map[label] for label in risk_data.index]

# Explode: Saare suspicious (non-unique) parts ko thoda bahar nikalna
explode = [0.2 if 'Unique' not in label else 0 for label in risk_data.index]

ax1.pie(risk_data, labels=risk_data.index, autopct='%1.2f%%',
        startangle=140, colors=colors, explode=explode, shadow=True,
        pctdistance=0.85, textprops={'fontsize': 10, 'fontweight': 'bold'})
ax1.set_title('KYB Analysis: Address Clustering Risk', fontsize=16, fontweight

# --- CHART 2: BAR CHART (Top 10 Shell Hubs) ---
top_10 = full_results['address'].value_counts().head(10)
sns.barplot(x=top_10.values, y=top_10.index, ax=ax2, palette='Reds_r', hue=top

ax2.set_title('Top 10 Suspicious Address Hubs', fontsize=16, fontweight='bold'
ax2.set_xlabel('Total Companies Registered', fontsize=12)
ax2.grid(axis='x', linestyle='--', alpha=0.3)

plt.tight_layout(pad=5.0)
plt.show()

# Verification
print("Risk Breakdown (Exact Numbers):")
print(risk_data)
```




```
Risk Breakdown (Exact Numbers):
risk_level
Extreme Cluster (>500)    126535
Unique Address (1)         97056
Multi-Entity (2-50)        61098
High Cluster (51-500)      52262
Name: count, dtype: int64
```

# Why we need neo4j or graph analytics when we can do in numpy and pandas

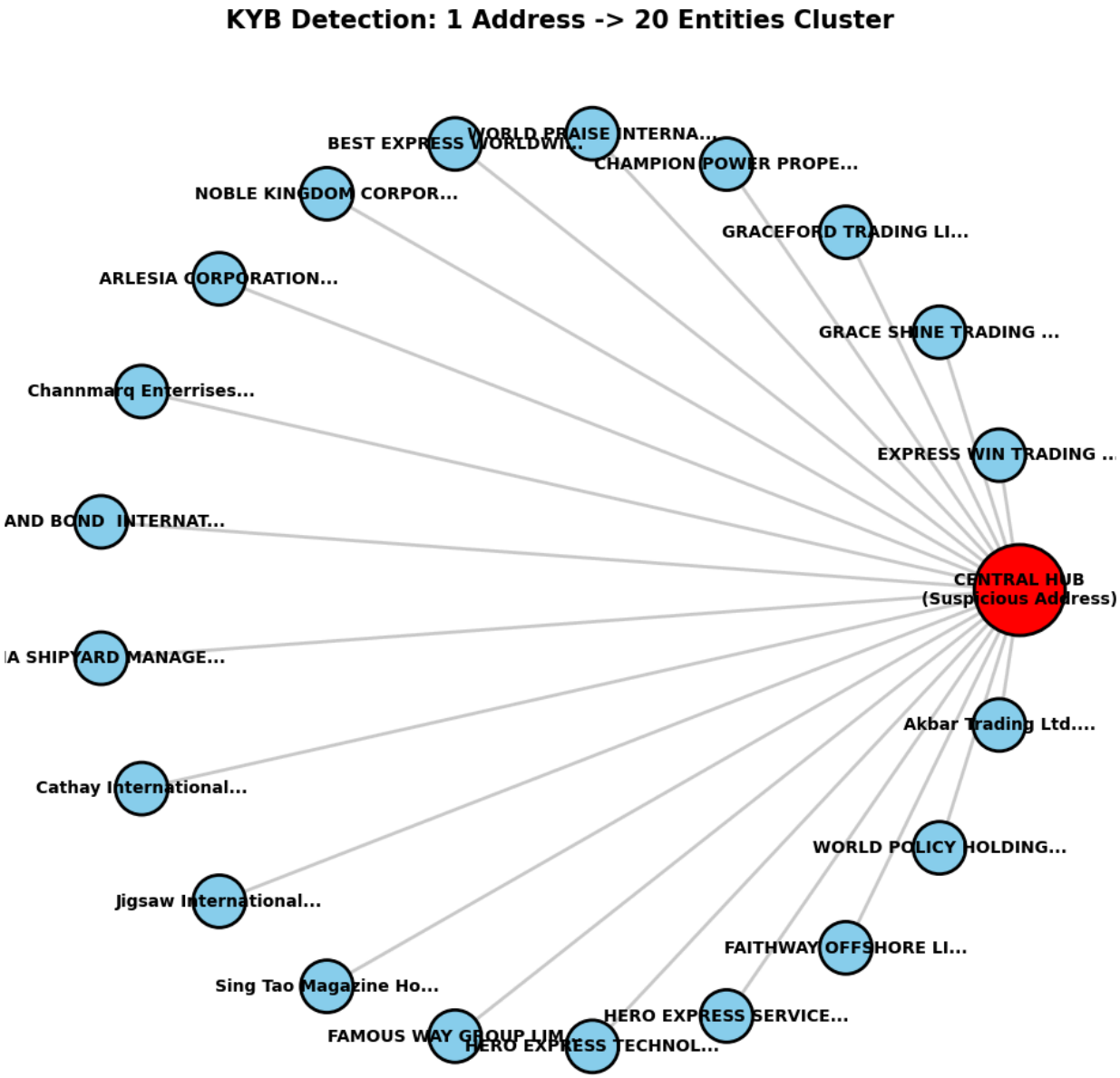
Iska jawab aapke Research Project (KYB & Shell Company Detection) ke "Depth" mein chhupa hai. Pandas tab tak achha hai jab tak aap "Flat" analysis kar rahe hain, lekin Neo4j tab zaroori ho jata hai jab aapko "Hidden Connections" dhoondne hon.

### Pandas vs Neo4j (KYB Perspective)

| Feature | Pandas (Jo aap abhi kar rahe hain) | Neo4j (Graph Database) |
|---|---|---|
| **Logic** | Row aur Column mein data dhundna. | Rishton (Relationships) ka jaal dhundna. |
| **Simple Search** | "Is address par kitni companies hain?" (**Aasaan**) | "Is address par kitni companies hain?" (Aasaan) |
| **Deep Link** | "Company A ka director, Company B ke director ka bhai hai?" (**Bahut Mushkil**) | "Find all paths between Entity A and Entity B." (**Bahut Fast**) |
| **Circular Link** | Ye detect karna ki Paisa A -> B -> C -> A ghoom raha hai. (**Impossible**) | Circular loops ko 1 second mein pakad leta hai. (**Specialty**) |
| **Scalability** | 1.85 lakh rows mein laptop hang hone lagta hai. | Karodon connections ko bina hang hue handle karta hai. |

In [19]:
```
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt

# 1. Sirf TOP address ki pehli 15-20 companies lo (Zyada mat lena warna mess b
top_address = full_results['address'].value_counts().idxmax()
subset = full_results[full_results['address'] == top_address].head(20)

# 2. Graph taiyar karna
G = nx.Graph()
addr_node = "CENTRAL HUB\n(Suspicious Address)"
G.add_node(addr_node, color='red', size=3000)

for _, row in subset.iterrows():
    company_label = row['name'][:20] + "..." # Naam chhote rakho taaki readabl
    G.add_node(company_label, color='skyblue', size=1000)
```

```
    G.add_edge(company_label, addr_node)

# 3. Plotting (Circular Layout is best for your report)
plt.figure(figsize=(12, 12))
pos = nx.circular_layout(G) # Sabko ek circle mein arrange karne ke liye

# Draw Nodes
node_colors = [nx.get_node_attributes(G, 'color')[n] for n in G]
node_sizes = [nx.get_node_attributes(G, 'size')[n] for n in G]

nx.draw_networkx_nodes(G, pos, node_size=node_sizes, node_color=node_colors, e
nx.draw_networkx_edges(G, pos, width=2, alpha=0.4, edge_color='gray')
nx.draw_networkx_labels(G, pos, font_size=10, font_weight='bold')

plt.title(f"KYB Detection: 1 Address -> {len(subset)} Entities Cluster", fonts
plt.axis('off')
plt.show()
```

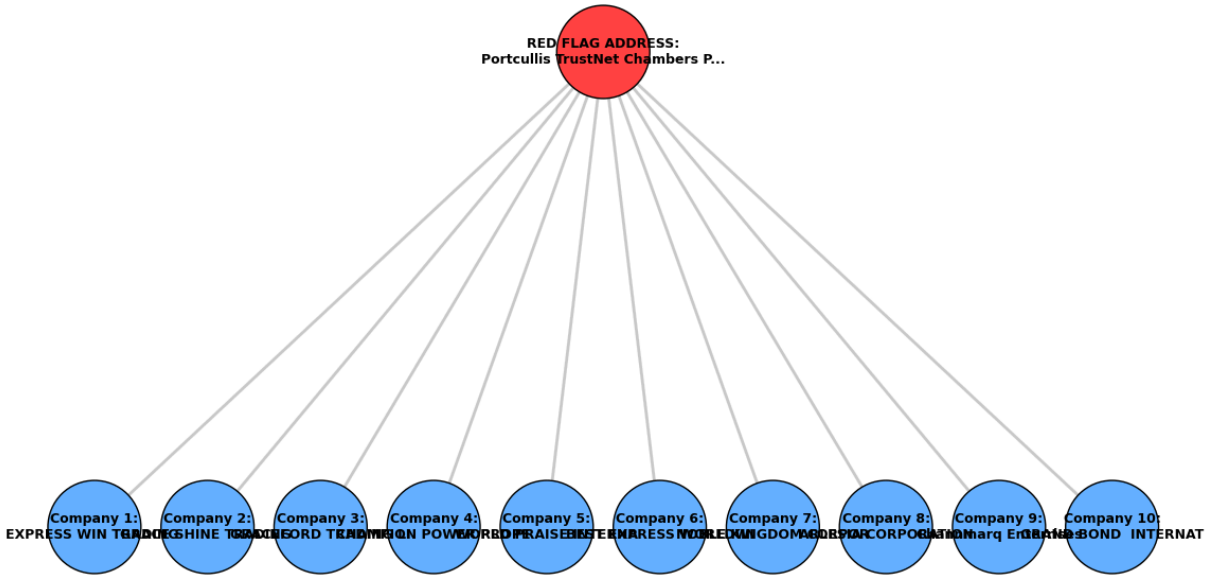



In [20]:

```
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt

# 1. Sabse bada suspicious address nikalna
top_address = full_results['address'].value_counts().idxmax()
# Sirf top 10 companies lo taaki ekdum saaf dikhe
subset = full_results[full_results['address'] == top_address].head(10)

# 2. Graph taiyar karna
G = nx.Graph()
addr_label = f"RED FLAG ADDRESS:\n{top_address[:30]}..."
G.add_node(addr_label, level=0)

for i, row in subset.iterrows():
```

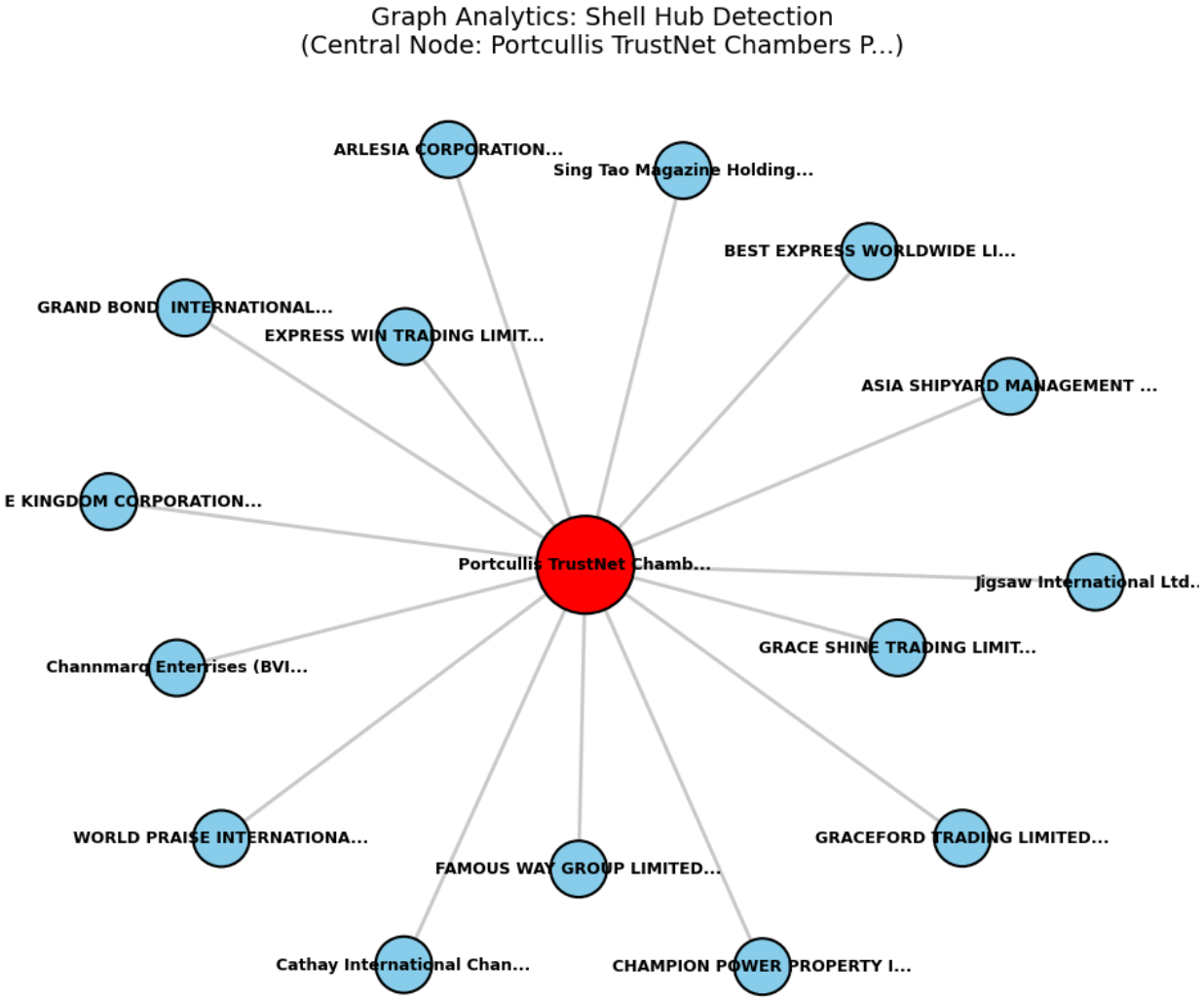
```
    comp_label = f"Company {i+1}:\n{row['name'][:20]}"
    G.add_node(comp_label, level=1)
    G.add_edge(addr_label, comp_label)

# 3. Tree-like Position (Manual Layout for Clarity)
pos = {}
pos[addr_label] = (0.5, 0.9) # Address sabse upar center mein

for i, node in enumerate(list(G.nodes())[1:]):
    pos[node] = (i / (len(subset) - 1), 0.1) # Companies niche ek line mein

# 4. Plotting
plt.figure(figsize=(15, 8))

# Draw Nodes
nx.draw_networkx_nodes(G, pos, node_size=4000, node_color=['#ff4444' if n == a
nx.draw_networkx_edges(G, pos, width=2, alpha=0.4, edge_color='gray')

# Draw Labels (Address upar, Companies niche)
nx.draw_networkx_labels(G, pos, font_size=9, font_weight='bold')

plt.title("KYB Detection: Hierarchical Relationship View", fontsize=16, fontwe
plt.ylim(0, 1.1) # Thoda space upar niche
plt.axis('off')
plt.show()
```

**KYB Detection: Hierarchical Relationship View**




In [23]:
```
import networkx as nx
import matplotlib.pyplot as plt

# 1. Sabse bada suspicious address aur uski companies ko select karna
top_address = full_results['address'].value_counts().idxmax()
# Sirf top 15 companies lena (zyada lenge toh visual mess ban jayega)
```

```
subset = full_results[full_results['address'] == top_address].head(15)

# 2. Graph Object banana
G = nx.Graph()

# 3. Nodes aur Edges add karna
# Address ko 'Hub' node banana
G.add_node(top_address, type='Address', color='red', size=3000)

for _, row in subset.iterrows():
    G.add_node(row['name'], type='Entity', color='skyblue', size=1000)
    G.add_edge(row['name'], top_address)

# 4. Professional Visualisation (Spring Layout)
plt.figure(figsize=(12, 10))
pos = nx.spring_layout(G, k=0.6, iterations=50) # Nodes ko balance karne ke li

# Draw Nodes
colors = [nx.get_node_attributes(G, 'color')[n] for n in G]
sizes = [nx.get_node_attributes(G, 'size')[n] for n in G]

nx.draw_networkx_nodes(G, pos, node_size=sizes, node_color=colors, edgecolors=
nx.draw_networkx_edges(G, pos, width=2, alpha=0.4, edge_color='gray')

# Labeling (Sirf Hub aur kuch entities ke liye)
labels = {n: n[:25] + '...' for n in G.nodes()}
nx.draw_networkx_labels(G, pos, labels, font_size=9, font_weight='bold')

plt.title(f"Graph Analytics: Shell Hub Detection\n(Central Node: {top_address[
plt.axis('off')
plt.show()
```




# Agar aap Neo4j use karte, toh aap yeh query bol sakte the:

MATCH (a:Address)<-[:REGISTERED_AT]-(e:Entity) RETURN a, count(e) AS density ORDER BY density DESC Yeh query 1.85 lakh rows ko 1 second se bhi kam mein scan karke bata deti hai ki kahan clustering ho rahi hai.

In [ ]: